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Automated Platforms for Reaction Optimization: Merging Self-Driving Labs with Flow Photochemistry

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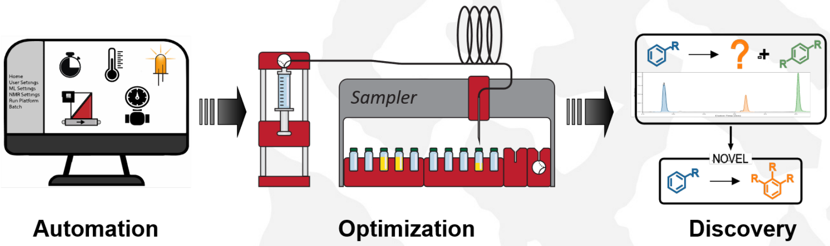
Abstract

Self-driving laboratories (SDLs) are emerging as powerful tools to accelerate reaction discovery and optimization in organic chemistry.[1] By combining automation, advanced algorithms, and real-time feedback, SDLs can efficiently navigate complex chemical spaces and uncover high-performance reaction conditions. Despite their potential, widespread use has been limited by cost, infrastructure demands, and accessibility. Recent efforts show that modular, customizable, and affordable SDLs can overcome these barriers, making automation feasible even in resource-limited settings.[2] Such approaches not only democratize access to cutting-edge experimentation but also expand opportunities for innovation across diverse catalytic and synthetic transformations.

Extending this concept, we applied automation to gas-phase chemistry, using a photoreactor to optimize the light-driven reduction of CO_2 to CO under reproducible and scalable conditions.[3,4] The on-demand CO was then directly used in carbonylative cross-coupling reactions, linking sustainable feedstock valorization with transformations of clear synthetic relevance. This example highlights how automated platforms can accelerate optimization while connecting new catalytic methodologies with practical applications in organic synthesis.



References

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